

The Conscious/Unconscious Distinction Matters, But Consciousness Theories Don't: Emergent Cooperation in a Thermodynamic Multi-Agent Engine

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Abstract

We present Symtropy, an open-source multi-agent simulation engine where a consciousness-inspired integration variable (Φ) modulates rigid body dynamics through five coupling channels. Across 1,800+ simulation runs with 8 experimental conditions, we establish three findings: (1) Thermodynamic enforcement combined with free energy gradient descent produces spatial clustering, with agents forming groups 35% tighter than unconscious agents ($\Phi=0$). (2) A six-condition ablation shows the FEP gradient is the clustering mechanism and epistemic offloading is the survival mechanism—neither alone is sufficient. (3) Critically, the *specific* consciousness metric is interchangeable: IIT-inspired Φ , Shannon entropy, and constant scalars produce statistically identical clustering (5.57 ± 0.48 vs 5.65 ± 0.83 vs 5.57 ± 0.48 , 20 seeds). Only the $\Phi=0$ condition differs (7.50 ± 3.41). This demonstrates that the conscious/unconscious distinction matters for social organization, but the choice between consciousness theories does not. Cooperation scales with population (55% survival at $N=12$, 92% at $N=96$), and a phase diagram shows population density, not energy cost, is the critical factor. The engine supports N -dimensional physics (2D/3D/4D) with 102ns GJK collision detection. Code: AGPL-3.0, 230+ tests.

Keywords: emergent cooperation, thermodynamic enforcement, consciousness metrics, free energy principle, multi-agent systems, ablation study

1 Introduction

The relationship between consciousness and multi-agent social organization remains poorly understood. Integrated Information Theory (?) proposes consciousness is identical to integrated information (Φ), while the Free Energy Principle (?) proposes that conscious systems minimize variational free energy. Both frameworks are computationally formal, yet neither has been tested as a real-time modulator of physics in multi-agent simulation.

We ask three questions: (1) Does energy-costly information processing produce spatial cooperation? (2) Which mechanism drives clustering—the gradient, the cost reduction, or both? (3) Does the specific consciousness metric matter?

We present Symtropy, a consciousness-physics engine where Φ modulates rigid body dynamics through five coupling channels. Under strict energy conservation with a Landauer-bounded floor, agents that minimize variational free energy form spatial clusters. Critically, we show this clustering is invariant to the

choice of consciousness metric but sensitive to the conscious/unconscious distinction.

1.1 Contributions

1. Five-channel consciousness-physics coupling model (Section 3)
2. Thermodynamic enforcement with epistemic offloading (Section 4)
3. Six-condition ablation disentangling clustering and survival mechanisms
4. Metric independence: Φ , entropy, random, and constant produce identical behavior
5. Causal test: $\Phi=0$ clusters 35% less when wired into the gradient
6. Scaling and phase diagram showing cooperation is robust across regimes

2 Related Work

IIT has been implemented computationally (?) but exclusively as a measurement tool. Active inference agents (??) operate in fixed environments. Tierra (?) demonstrated energy-gated ecosystems; Flow Lenia (?) showed mass conservation enables coexistence. No existing physics engine supports consciousness-coupled dynamics.

Our mechanism is closest to Nowak’s network reciprocity (?) but the network structure *emerges* from thermodynamics rather than being imposed. This parallels Prigogine’s dissipative structures (?): the pattern depends on the physics, not the medium.

3 Consciousness-Physics Coupling

An agent i at time t has state $S_i = (x_i, v_i, \omega_i, \Phi_i, E_i, h_i, \sigma_i, \pi_i)$ where $x_i \in R^D$ is position, $\Phi_i \in [0, 1]$ is the integration variable, E_i is energy, $h_i \in [0, 1]^8$ is the harmony activation vector, σ_i is prediction error, and $\pi_i = 1/(1 + \sigma_i)$ is motor precision.

Channel 1: Force Modulation. Applied force scales by $G_i = g(\Phi_i) \times \pi_i$ where g is a 4-tier safety function.

Channel 2: Energy Budget. Energy depletes through movement ($c_{\text{move}} \times |v|$), consciousness maintenance ($c_{\text{maint}} \times (1 + 0.5\Phi)$), and collisions ($c_{\text{coll}} \times |J_n|$).

Channel 3: Sanctuary Zones. When Sacred Stillness harmony $h_7 > 0.6$, a zone dampens collision impulses by up to 90%.

Channel 4: Harmony Fields. Each agent emits a harmony field with $1/r^2$ falloff, inspired by McFadden’s CEMI theory (?). Friction modulates as $\mu_{\text{eff}} = \mu(1 - 0.5R)$ where R is the resonance (cosine similarity of harmony vectors).

Channel 5: Prediction Error Feedback. On collision, prediction error spikes ($\Delta\sigma = \min(0.01|J|, 2.0)$), reducing motor precision temporarily. Based on?: motor commands are proprioceptive predictions.

4 Thermodynamic Enforcement

Energy is strictly conserved. Cooperation reduces costs through **epistemic offloading**: when agents with resonance $R > 0.5$ are within range, prediction error decays 10% faster and maintenance cost is refunded by $50\% \times (R - 0.5) \times 2$. This is grounded in Landauer’s principle (?): processing information has a minimum thermodynamic cost of $k_B T \ln 2$ per bit.

Cooperation does NOT generate energy. It reduces expenditure. Solo survival: ~ 4 minutes. With offload-

ing: ~ 6 minutes. Only energy wells provide indefinite survival.

5 Experiments (1,800+ Runs)

5.1 Ablation Study

Six conditions, 12 agents, 5000 ticks, 5 seeds, 2 energy wells (Table ??).

Finding 1: The FEP gradient is the clustering mechanism ($15.1 \rightarrow 1.8$). Epistemic offloading is the survival mechanism ($72\% \rightarrow 45\%$ collapse). Neither alone is sufficient.

5.2 Metric Independence

Five consciousness metrics, 30 seeds each, Φ not coupled into the gradient (Table ??).

Finding 2: All five metrics produce statistically identical clustering. The specific consciousness metric is interchangeable.

5.3 Causal Test: Φ Wired Into the Gradient

We coupled Φ into three behavioral pathways: cooperation urgency ($0.5 + \Phi$), resonance gating ($0.3 + \Phi \times 0.7$), and danger sensitivity ($0.5 - \Phi \times 0.4$). Six conditions, 24 agents, 20 seeds (Table ??).

Finding 3 (central result): When wired into the gradient, $\Phi > 0$ produces 35% tighter clustering than $\Phi = 0$ (5.57 vs 7.50). But Φ , entropy, and constant 0.5 are interchangeable (~ 5.6). The conscious/unconscious distinction matters; the choice between theories does not.

Finding 4: Unconscious agents survive 5% longer (2993 vs 2859). Consciousness amplifies social drive at a survival cost.

5.4 Scaling and Phase Diagram

Cooperation scales: 55% survival at $N=12$, 92% at $N=96$ (20 seeds per condition). Phase diagram (8 costs $\times$ 4 densities, 10 seeds): above 24 agents, cooperation emerges at all tested energy costs (0.02 – 0.50 J/tick).

Finding 5: Population density, not energy cost, is the critical factor for cooperation.

6 Discussion

6.1 Limitations

Φ is a softmax bottleneck approximation, not full IIT partition analysis. The FEP gradient uses handcrafted heuristics, not learned policies. The $O(n^2)$ broadphase limits scale to ~ 500 bodies. Agents do not learn or adapt strategies.

6.2 Implications

Our metric independence result is a universality finding analogous to Prigogine’s dissipative structures (?) and Reynolds’ flocking (?): the macro-pattern depends on the thermodynamic cost structure, not the internal state metric. The conscious/unconscious finding adds nuance: having *some* nonzero integration variable amplifies social behavior, but which theory generates it is irrelevant.

This aligns with the “behavioral inference principle” for machine consciousness: behavioral equivalence holds across radically different internal metrics, but the presence/absence of integration creates a measurable behavioral difference.

7 Conclusion

Across 1,800+ runs, we show: (1) clustering requires both gradient descent and cost reduction; (2) the specific consciousness metric is interchangeable; (3) the conscious/unconscious distinction produces a 35% clustering difference when wired into behavioral pathways. Consciousness amplifies social drive at a survival cost—a trade-off that may be fundamental to systems where information integration is thermodynamically expensive.

The Symtropy engine and all experiments are open-source. Results are reproducible via `cargo run -example`.

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